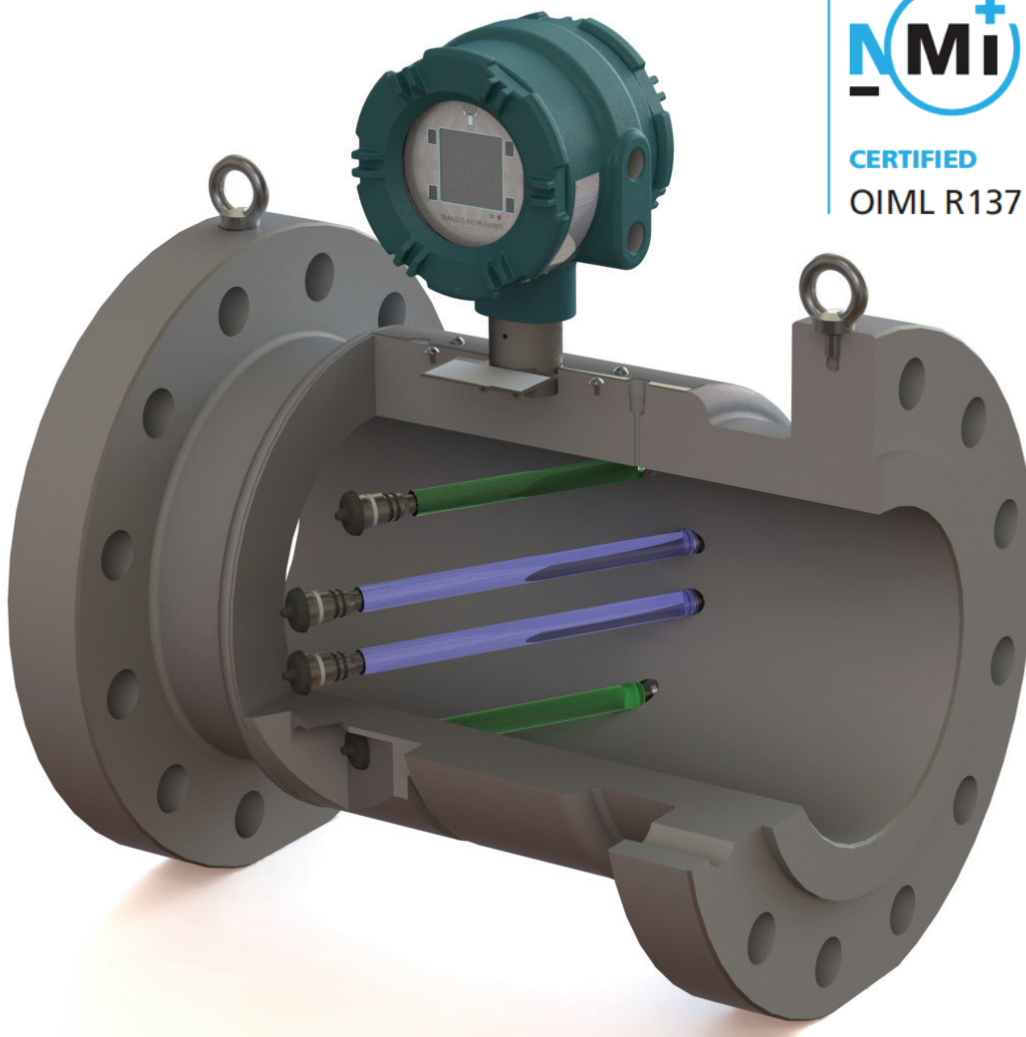




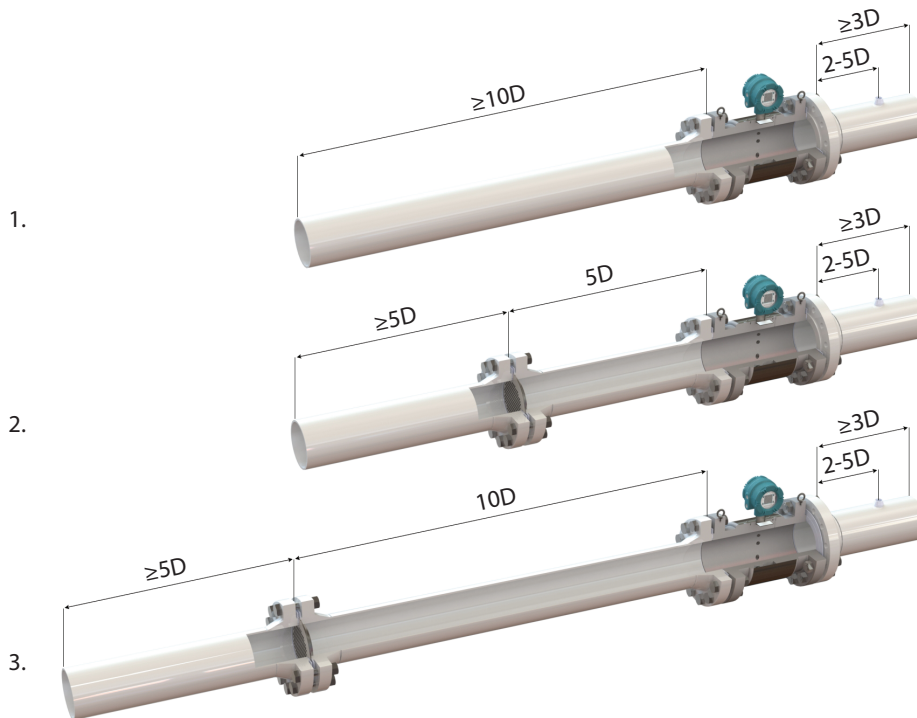
# Technical Datasheet

**UIM-4F**  
Metric

UIM Series Flowmeter

|                                  |                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Principle of operation           | Broadband continuous wave – transit time                                                                                                                                                                                                                                                                              |
| Sizes                            | 150, 300, 600 and 900lbs; 3 – 30", other sizes on request                                                                                                                                                                                                                                                             |
| Flange type                      | ANSI, DIN, others on request                                                                                                                                                                                                                                                                                          |
| Pressure ranges                  | Up to 153 bar (2250 psi)                                                                                                                                                                                                                                                                                              |
| Ambient temperature              | -40 to +60°C (-40 to +140°F)                                                                                                                                                                                                                                                                                          |
| Process temperature              | -30 to +80°C (-22 to +176°F)                                                                                                                                                                                                                                                                                          |
| Typical uncertainty              | 0.5% ( $Q_t - Q_{max}$ ) factory calibration<br>1% ( $Q_{min} - Q_t$ ) factory calibration<br><br>0.2% ( $Q_t - Q_{max}$ ) flow calibration<br>0.5% ( $Q_{min} - Q_t$ ) flow calibration                                                                                                                              |
| Metrology                        | AGA-9 compliant<br>OIML R137-1&2<br>MID 2014/32/EU                                                                                                                                                                                                                                                                    |
| Repeatability                    | 0.1%                                                                                                                                                                                                                                                                                                                  |
| Turndown                         | 100:1 (pipe size dependent)                                                                                                                                                                                                                                                                                           |
| Meter body materials             | Carbon steel ASTM A350-LF2 Cl.1<br>Stainless steel ASTM A182-F316<br>Other materials on request                                                                                                                                                                                                                       |
| Transducers                      | All metal Titanium; retractable version available as option                                                                                                                                                                                                                                                           |
| Pressure port                    | 1/4" NPT female, others on request                                                                                                                                                                                                                                                                                    |
| Electronic enclosure material    | Epoxy painted, low copper aluminum alloy                                                                                                                                                                                                                                                                              |
| Electronic enclosure cable entry | M20x1.5 female<br>1/2" NPT female                                                                                                                                                                                                                                                                                     |
| Power supply                     | Main power: 14 - 28.8V, 670mW max<br><br>I/O option board power: 10.8 - 28.8V, 225mW max                                                                                                                                                                                                                              |
| User interface                   | 128x128 dot matrix LC Display, 4 keys                                                                                                                                                                                                                                                                                 |
| Interface ports                  | 1x USB (not intrinsically safe)<br>1x HF Frequency output / LF pulse output                                                                                                                                                                                                                                           |
| Optional interface ports         | <u>I/O Option Board</u><br>1x RS485, two wire, externally powered<br>2x Digital, software configurable (HF, LF, status)<br>1 x 4-20mA loop powered output (pending)<br><u>PT Option board</u><br>Pressure and temperature sensors<br><u>4-20mA/HART Option Board</u><br>1 x 4-20mA loop powered output (HART pending) |
| Communication protocols          | MODBUS (RS485 and USB)                                                                                                                                                                                                                                                                                                |
| Hazardous area certification     | ATEX EX II 1 G Ex ia IIC T4 Ga, Zone 0<br>IECEX Ex ia IIC T4 Ga<br>CSA/FM Class I, Division 1, Group A,B,C,D T4                                                                                                                                                                                                       |
| Ingress Protection               | IP66, NEMA 4X                                                                                                                                                                                                                                                                                                         |

|                     | Nominal Size | Schedule | Internal diameter [mm] | Flow [m <sup>3</sup> /hr] |      |      | Turndown |
|---------------------|--------------|----------|------------------------|---------------------------|------|------|----------|
|                     |              |          |                        | Qmax                      | Qt   | Qmin |          |
| Flowranges [Metric] | 3" / DN80    | 40       | 77.9                   | 610                       | 61   | 7    | 88       |
|                     |              | 80       | 73.7                   | 540                       | 54   | 7    | 78       |
|                     | 4" / DN100   | 40       | 102.3                  | 1000                      | 100  | 12   | 84       |
|                     |              | 80       | 97.2                   | 900                       | 90   | 11   | 82       |
|                     | 6" / DN150   | 40       | 154.1                  | 2020                      | 202  | 21   | 97       |
|                     |              | 80       | 146.3                  | 1820                      | 182  | 19   | 96       |
|                     | 8" / DN200   | 40       | 202.7                  | 3490                      | 349  | 35   | 100      |
|                     |              | 80       | 193.7                  | 3190                      | 319  | 32   | 100      |
|                     | 10" / DN250  | 40       | 254.5                  | 5500                      | 550  | 55   | 100      |
|                     |              | 80       | 242.9                  | 5100                      | 510  | 51   | 100      |
|                     | 12" / DN300  | STD      | 304.7                  | 7900                      | 790  | 79   | 100      |
|                     |              | 80       | 289.0                  | 7100                      | 710  | 71   | 100      |
|                     | 14" / DN350  | STD      | 336.5                  | 9700                      | 970  | 97   | 100      |
|                     |              | 80       | 317.5                  | 8600                      | 860  | 86   | 100      |
|                     | 16" / DN400  | STD      | 387.3                  | 12800                     | 1280 | 128  | 100      |
|                     |              | 80       | 363.5                  | 11300                     | 1130 | 113  | 100      |
|                     | 18" / DN450  | STD      | 437.9                  | 16300                     | 1630 | 163  | 100      |
|                     |              | 80       | 409.3                  | 14300                     | 1430 | 143  | 100      |
|                     | 20" / DN500  | XS       | 482.6                  | 19800                     | 1980 | 198  | 100      |
|                     |              | 80       | 455.6                  | 17700                     | 1770 | 177  | 100      |
|                     | 24" / DN600  | XS       | 584.6                  | 28100                     | 2810 | 290  | 97       |
|                     |              | 80       | 547.7                  | 24600                     | 2460 | 255  | 97       |
|                     | 30" / DN750  | 30       | 730.2                  | 42300                     | 4230 | 453  | 94       |
|                     |              | wt = 35  | 692.0                  | 38000                     | 3800 | 407  | 94       |



Configuration 1. – Recommended configuration without flow conditioner

Configuration 2. – Recommended configuration with flow conditioner

Configuration 3. – Conservative configuration with flow conditioner

# Flowranges and configurations

# Dimensions and weights

| Dimensions and weights [Metric] | Nominal Size | Rating | A- Length [mm] | B- Width [mm] | C - Height [mm] | Weight [kg] |
|---------------------------------|--------------|--------|----------------|---------------|-----------------|-------------|
|                                 | 3" / DN80    | 150    | 300            | 190           | 372             | 35          |
|                                 |              | 300    | 350            | 210           | 382             | 36          |
|                                 |              | 600    | 350            | 210           | 382             | 38          |
|                                 | 4" / DN100   | 150    | 350            | 230           | 410             | 56          |
|                                 |              | 300    | 350            | 255           | 422             | 60          |
|                                 |              | 600    | 400            | 275           | 447             | 81          |
|                                 | 6" / DN150   | 150    | 350            | 280           | 457             | 57          |
|                                 |              | 300    | 400            | 320           | 477             | 96          |
|                                 |              | 600    | 400            | 355           | 510             | 120         |
|                                 | 8" / DN200   | 150    | 450            | 345           | 515             | 145         |
|                                 |              | 300    | 500            | 380           | 532             | 175         |
|                                 |              | 600    | 500            | 420           | 570             | 192         |
|                                 | 10" / DN250  | 300    | 550            | 445           | 610             | 245         |
|                                 |              | 600    | 550            | 510           | 640             | 295         |
|                                 | 12" / DN300  | 300    | 600            | 520           | 675             | 285         |
|                                 |              | 600    | 600            | 560           | 695             | 365         |
|                                 | 14" / DN350  | 300    | 600            | 585           | 705             | 405         |
|                                 |              | 600    | 600            | 605           | 730             | 435         |
|                                 | 16" / DN400  | 300    | 700            | 650           | 755             | 620         |
|                                 |              | 600    | 700            | 690           | 775             | 660         |
|                                 | 18" / DN450  | 600    | 900            | 745           | 905             | 950         |
|                                 | 20" / DN500  | 600    | 1000           | 815           | 940             | 1150        |
|                                 | 24" / DN600  | 600    | 1100           | 940           | 1010            | 1500        |
|                                 | 30" / DN750  | 600    | 1300           | 1130          | 1220            | 2450        |

